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Inverse Problems and Imaging (02624)

Week 6

Parameter choice rules

Consider

$$Kx = y^\delta = Kx^* + n.$$

For Tikhonov regularization

$$x_\alpha = R_\alpha y^\delta = \sum_j \frac{\mu_j}{\mu_j^2 + \alpha} (y^\delta, y_j) x_j$$

we have

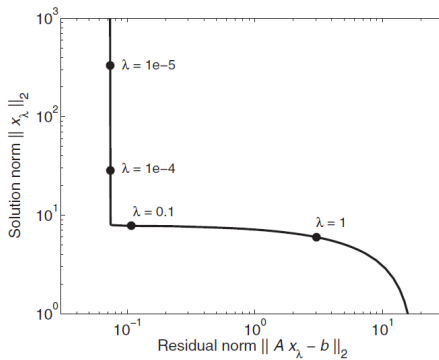
$$\begin{aligned} x^* - x_\alpha &= (x^* - R_\alpha Kx^*) - R_\alpha n \\ &= \sum_j \frac{\alpha}{\mu_j^2 + \alpha} (x^*, x_j) x_j - \sum_j \frac{\mu_j}{\mu_j^2 + \alpha} (n, y_j) x_j. \end{aligned}$$

Regularization bias vs. perturbation error

The choice of optimal regularization parameter is always a challenge.

L-curve: $\alpha \mapsto (\log F_1(\alpha), \log F_2(\alpha))$

$$F_1(\alpha) = \|Kx_\alpha - y_\delta\| \quad F_2(\alpha) = \|x_\alpha\|$$



Discrepancy principle

Theorem: There is a unique α such that $F_1(\alpha) = \|Kx_\alpha - y_\delta\| = \delta$.

Proof:

- Show that $\lim_{\alpha \rightarrow 0} F_1(\alpha) = \|P^\perp y^\delta\| < \delta$.
- Show that F_1 is continuous and strictly increasing.
- Show that $\lim_{\alpha \rightarrow \infty} F_1(\alpha) = \|y^\delta\| > \delta$.

Choose $\alpha = \alpha^*$ so that the residual meets the noise level

$$F_1(\alpha^*) = \delta.$$

is called Mrozov's discrepancy principle. For better stability, choose safety factor $\nu > 1$ to solve

$$F_1(\alpha) = \nu\delta$$

This parameter choice rule is admissible: $\lim_{\delta \rightarrow 0} x_\alpha = x^*$.

L-curve criterion

Choose the parameter in the corner of the L-curve - the point of maximal curvature.

Formula for $\hat{F}_j = 2 \log F_j$

$$c_\alpha = 2 \frac{\hat{F}'_1 \hat{F}''_2 - \hat{F}'_2 \hat{F}''_1}{((\hat{F}'_1)^2 + (\hat{F}'_2)^2)^{3/2}}$$

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NB: The L-curve criterion only depends on y_δ and not on $\|e\| = \delta$.

Bad news: Such a parameter choice rule is, in general, not admissible.

Rules for TSVD

Discrepancy:

Choose $k = k_{\text{dp}}$ such that $\|Ax_k - y_\delta\| \geq \nu\delta > \|Ax_{k+1} - y_\delta\|$.

L-curve:

Choose $k = k_l$ near the “overall corner” of the curve.

Numerical differentiation

Matlab!